

ANKARA BOLGE, Turkey

WMO: 171300

Lat: 39.973N

Lon: 32.863E

Elev: 2923

StdP: 13.21

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	21.6	8.6	9.5	25.5	12.8	11.7	28.5	14.4	40.3	12.8	39.9	4.3	50	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.5	94.6	62.6	91.6	62.3	89.0	61.8	66.5	85.0	65.2	83.8	64.0	82.6	5.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	88.6	72.2	59.0	83.1	71.2	57.4	78.5	70.4	32.7	84.7	31.7	83.7	30.8	82.3	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.4	12.3	10.9	10.8	99.6	5.2	3.0	7.1	101.7	4.1	103.5	1.2	105.1	-2.6	107.3		
			9.7	68.9	4.8	2.1	6.2	70.5	3.4	71.7	0.7	72.9	-2.9	74.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	35.9	39.0	45.3	54.5	62.5	69.8	75.8	76.7	68.4	57.6	47.4	38.3	
	DBStd	15.57	7.52	8.06	7.16	6.55	5.82	5.02	4.60	4.57	6.66	6.59	6.46	7.24	
	HDD50	1460	438	314	177	26	0	0	0	0	0	15	124	366	
	HDD65	4332	903	729	611	322	120	13	0	0	42	238	527	827	
	CDD50	3660	1	5	31	160	387	595	801	826	551	252	47	4	
	CDD65	1057	0	0	0	6	42	158	336	361	144	10	0	0	
	CDH74	11566	0	0	2	94	458	1596	3796	3974	1522	124	0	0	
	CDH80	4695	0	0	0	7	84	528	1694	1815	552	15	0	0	
Wind	WSAvg	4.6	4.1	4.5	4.5	4.7	4.7	5.0	5.9	6.3	4.9	3.8	3.4	3.7	
Precipitation	PrecAvg	16.2	1.7	1.4	1.7	1.8	2.1	1.4	0.5	0.5	0.7	1.2	1.3	1.8	
	PrecMax	24.4	4.0	3.4	3.8	4.4	4.8	4.8	2.6	3.2	3.1	6.8	4.4	4.6	
	PrecMin	9.5	0.2	0.1	0.4	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	3.3	1.0	0.9	0.9	1.0	1.1	1.0	0.5	0.7	0.8	1.1	0.9	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	63.8	70.9	81.0	85.9	92.1	99.5	98.7	93.5	81.6	70.4	59.9	
		MCWB	46.6	48.6	50.1	55.4	58.1	62.6	63.1	62.6	59.9	55.5	52.3	48.5	
	2%	DB	51.3	58.4	65.4	76.8	82.3	88.8	94.7	94.9	89.6	77.3	67.2	54.2	
		MCWB	43.8	46.3	48.3	53.7	58.0	61.7	63.0	62.8	60.1	55.9	51.8	46.5	
	5%	DB	48.3	54.1	61.9	72.5	79.3	85.8	91.5	92.1	86.6	73.9	62.9	51.4	
		MCWB	42.4	44.8	47.0	52.8	57.5	60.9	62.4	62.6	59.4	54.8	49.9	45.3	
	10%	DB	45.8	50.9	58.0	68.4	75.7	82.8	88.6	89.1	82.8	70.1	58.8	48.4	
		MCWB	41.4	43.3	45.9	51.2	56.7	60.5	62.2	62.2	58.3	54.1	48.1	43.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	50.9	52.0	57.1	61.8	65.6	68.3	68.9	64.3	60.1	56.3	50.3
			MCDB	54.3	61.9	64.8	75.8	76.7	83.8	87.1	85.1	83.0	67.1	65.6	56.5
2%		WB	44.7	47.6	49.8	55.4	60.2	64.1	66.5	66.7	62.3	58.3	52.8	48.0	
		MCDB	49.8	55.2	62.3	73.1	76.8	82.3	84.9	85.2	80.1	68.8	64.4	52.7	
5%		WB	43.1	45.7	48.2	53.9	59.0	62.9	65.1	65.5	61.1	57.1	50.4	46.0	
		MCDB	47.4	52.5	58.9	69.2	75.2	80.3	83.7	84.2	80.0	69.1	60.3	50.4	
10%		WB	41.6	43.9	46.6	52.3	57.7	61.8	64.0	64.3	60.0	55.9	48.9	44.3	
		MCDB	45.4	50.1	56.2	65.6	72.4	78.3	82.8	82.8	78.6	67.2	57.2	48.1	
Mean Daily Temperature Range	5% DB	MDBR	10.8	14.8	17.5	20.4	20.7	22.4	23.6	23.5	22.8	20.1	18.7	12.3	
		MCDBR	14.3	20.5	24.4	26.8	25.4	25.8	26.3	25.9	27.0	25.6	23.9	16.7	
		MCWBR	8.8	10.9	11.5	10.5	9.3	7.7	7.4	7.6	9.0	10.6	11.7	10.2	
	5% WB	MCDBR	12.7	18.0	20.2	23.5	22.8	23.2	24.0	23.3	23.4	19.4	20.8	14.6	
		MCWBR	8.1	9.8	10.1	9.8	9.0	7.8	7.7	7.5	8.1	8.5	10.6	9.0	
Clear-Sky Solar Irradiance	taub	0.304	0.330	0.373	0.409	0.389	0.369	0.367	0.365	0.345	0.346	0.315	0.305		
	taud	2.383	2.332	2.234	2.159	2.284	2.360	2.364	2.347	2.402	2.397	2.454	2.442		
	Ebn at Noon	272	280	278	274	282	287	287	285	284	271	267	263		
	Edh at Noon	27	33	40	46	42	39	38	38	34	30	25	23		
All-Sky Solar Radiation	RadAvg	594	914	1278	1637	1947	2217	2352	2095	1650	1126	778	527		
	RadStd	73	134	114	148	141	184	99	86	113	81	76	68		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page